Microeconomics I

Class 8 — Choice under uncertainty: expected utility and risk aversion

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21 October 2026

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Short quiz
10:10	Warm-up; two choices, by hands up
10:17	Lotteries, continuity, independence
10:28	Your two choices, revisited
10:31	Expected utility: form, theorem, uniqueness
10:39	Money lotteries and risk aversion
10:44	Certainty equivalent and risk premium
10:50	Break
11:00	Practice: five questions on Part I so far
11:44	Summary

Practicalities

- Problem set 2: handed in at the start of class, Wed 21 Oct; returned graded on Tue 27 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: always true, or not?

Only good 1 gets dearer, $p^0 \to p^1$. At p^0 she chose x^0, with utility u^0.

Fixed basket: $p^1 \cdot x^0 \,/\, p^0 \cdot x^0 = 1.3$. True cost-of-living index: $I = e(p^1, u^0) \,/\, e(p^0, u^0)$.

$I \leq 1.3$	$I \geq 1$
$I = 1.3$	good 1 inferior $\Rightarrow I > 1.3$

Class 6, slides 13, 22. Mas-Colell, Whinston and Green (MWG) Props. 3.E.1, 3.E.2, pp. 58–59.

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$I \leq 1.3$

true: x^0 still reaches u^0

$I \geq 1$

$I = 1.3$

good 1 inferior $\Rightarrow I > 1.3$

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$I \leq 1.3$

true: x^0 still reaches u^0

$I \geq 1$

true: no bundle got cheaper

$I = 1.3$

good 1 inferior $\Rightarrow I > 1.3$

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true: no bundle got cheaper

$I = 1.3$

false: lower if she substitutes

good 1 inferior $\Rightarrow I > 1.3$

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$I \leq 1.3$

true: x^0 still reaches u^0

$I \geq 1$

true: no bundle got cheaper

$I = 1.3$

false: lower if she substitutes

good 1 inferior $\Rightarrow I > 1.3$

false: still $I \leq 1.3$

The fixed basket never understates: $1 \leq I \leq 1.3$, normal or inferior.

Class 6, slides 13, 22. Mas-Colell, Whinston and Green (MWG) Props. 3.E.1, 3.E.2, pp. 58–59.

Two choices: which would you take?

Imagine the prizes are real and paid once. No right answer.

First choice

A: 1 million euros for sure

B: 10 %: 5 million euros
89 %: 1 million euros
1 %: nothing

Second choice

C: 11 %: 1 million euros
89 %: nothing

D: 10 %: 5 million euros
90 %: nothing

Hands up: A or B? Then: C or D?

Hypothetical prizes; the structure of Allais (1953), as in MWG Example 6.B.2, p. 179.

Lotteries and expected utility

MWG 6.B

A simple lottery is a list of probabilities: a point in a triangle

Simple lottery: $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$; p_n = probability of outcome n.

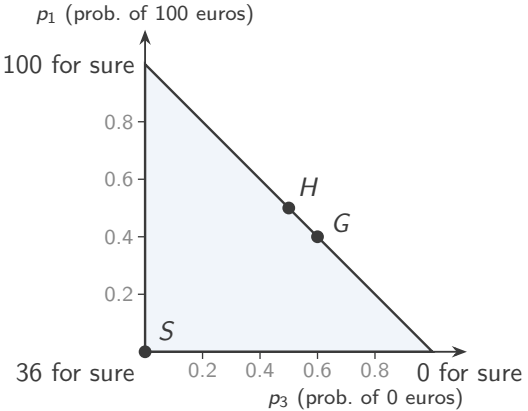
Outcomes: 100, 36, 0 euros.

$S = (0, 1, 0)$: 36 euros for sure

$H = (0.5, 0, 0.5)$: a coin, 100 or 0

$G = (0.4, 0, 0.6)$: 40 % chance of 100

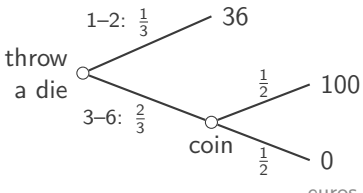
MWG Def. 6.B.1, p. 168; Fig. 6.B.1, p. 169 draws the same set as an equilateral triangle.



$p_2 = 1 - p_1 - p_3$: 0 on the hypotenuse, 1 at S

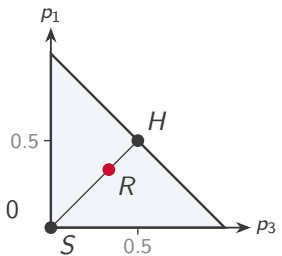
A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Compound lottery $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$: you get L_k with probability α_k.
Reduced lottery: $\alpha_1 L_1 + \cdots + \alpha_K L_K$.



$R = (S, H;\ \frac{1}{3}, \frac{2}{3})$

$\frac{1}{3}(0,1,0) + \frac{2}{3}(0.5,0,0.5)$
$= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
= one throw: 1–2 → 100,
3–4 → 36, 5–6 → 0



Only the reduced lottery matters (MWG's consequentialist premise): the die-then-coin tree and one throw are the same lottery.

MWG Def. 6.B.2, pp. 169–170; Figs. 6.B.2–6.B.3, p. 170.

Rational and continuous preferences over lotteries

Rational: complete, transitive (class 1).

Continuous: for all $L, L', L'' \in \mathcal{L}$ the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha \in [0,1] : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed.

Small probability changes never reverse a strict ranking.

> **Fails: safety first** (no chance of 0 beats any chance).
> $L = 100$, $L' = 0$, $L'' = 36$ euros for sure: second set $= [0, 1)$.

MWG Def. 6.B.3, p. 171 ("safety first": same page). Class 3: lexicographic order, not continuous.

Independence: mixing both with the same L'' keeps their ranking

Independence axiom: for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0,1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$



$\alpha L + (1-\alpha)L''$ $\qquad$ $\alpha L' + (1-\alpha)L''$

Tails gives both the same L'', and you get it instead of L or L', never with it: unlike bundles of goods.

MWG Def. 6.B.4, p. 171 (fn. 5: von Neumann and Morgenstern, 1944); coin-toss reading and Fig. 6.B.4, p. 172.

A with D breaks independence: the shared 89 % should not matter

	11 % of the time	89 % of the time
A	1 million euros	1 million euros
B	L'	1 million euros
C	1 million euros	0
D	L'	0

L': $\frac{10}{11}$ chance of 5 million euros, $\frac{1}{11}$ chance of 0 euros.

Allais (1953); MWG Example 6.B.2, pp. 179–180, and fn. 8: "roughly half the students choose this way". 81 experiments: "a fragile empirical finding" (Blavatskyy, Ortmann & Panchenko 2022, American Economic Journal: Microeconomics 14(1)).

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Independence with $\alpha = 0.11$: A $\succsim$ B $\iff$ 1 million for sure $\succsim L' \iff$ C $\succsim$ D.

Strictly preferring A and D breaks it.

Allais (1953); MWG Example 6.B.2, pp. 179–180, and fn. 8: "roughly half the students choose this way". 81 experiments: "a fragile empirical finding" (Blavatskyy, Ortmann & Panchenko 2022, American Economic Journal: Microeconomics 14(1)).

Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

U has the expected utility form if there are numbers $u_1, \ldots, u_N$ with $U(L) = u_1 p_1 + \cdots + u_N p_N$ for every L.

Such a U is a von Neumann–Morgenstern (vNM) expected utility function.

Her utilities for 100, 36, 0 euros: 10, 6, 0.

$U(S) = 6, \quad U(H) = 5$

$U(R) = \frac{1}{3} \cdot 10 + \frac{1}{3} \cdot 6 + \frac{1}{3} \cdot 0 = \frac{16}{3} = \frac{1}{3}\, U(S) + \frac{2}{3}\, U(H)$

MWG Def. 6.B.5 and Prop. 6.B.1 (expected utility form $\iff$ U linear), p. 173.

Continuity + independence ⇒ expected utility: straight, parallel lines

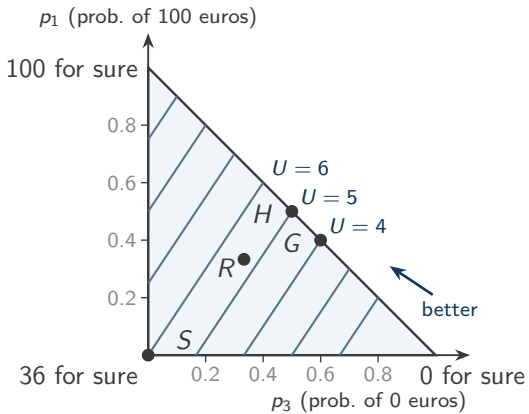
Expected utility theorem: if $\succsim$ on $\mathcal{L}$ is rational, continuous and satisfies independence, there are $u_1, \ldots, u_N$ such that for all L, L':

$$L \succsim L' \iff \sum_n u_n p_n \geq \sum_n u_n p'_n.$$

In the triangle: indifference curves are straight, parallel lines.

Conversely, every expected utility U satisfies independence (MWG Ex. 6.B.2).

MWG Prop. 6.B.3, (6.B.4), p. 176; proof pp. 176–178; Fig. 6.B.5, p. 175.



Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; u^2 flips her choice

utilities of (100, 36, 0 euros)	$U(G)$ $G = 40\,\%$ of 100	$U(S)$ $S = 36$ for sure	she picks
$u = (10, 6, 0)$	4	6	S

$\tilde{U}$ is another vNM utility for $\succsim$ $\iff$ $\tilde{U} = \beta U + \gamma$ with $\beta > 0$.

MWG Prop. 6.B.2, pp. 173–174.

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$2u + 3 = (23, 15, 3)$	11	15	S

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$u = (10, 6, 0)$	4	6	S
$2u + 3 = (23, 15, 3)$	11	15	S
$u^2 = (100, 36, 0)$	40	36	G

$\tilde{U}$ is another vNM utility for $\succsim \iff \tilde{U} = \beta U + \gamma$ with $\beta > 0$.

u^2 keeps the order of prizes, not lotteries:
unlike class 3, expected utility is cardinal.

MWG Prop. 6.B.2, pp. 173–174.

Money lotteries and risk aversion

MWG 6.C

A money lottery is a distribution function F; $U(F) = \int u(x)\, dF(x)$

$F(x)$ = probability that the payoff is at most x.

$U(F) = \int u(x)\, dF(x)$; with prizes x_n and probabilities p_n: $\sum_n p_n\, u(x_n)$.

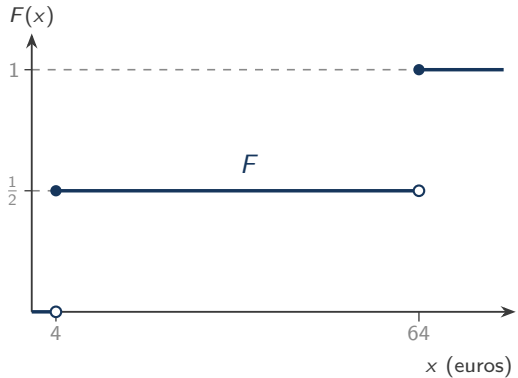
U: vNM utility, on lotteries.

u: Bernoulli utility, on money — increasing, continuous.

Her 10, 6, 0 for 100, 36, 0 euros: $u(x) = \sqrt{x}$.

MWG p. 183; (6.C.1) and fn. 12, p. 184; p. 185.

F: 4 or 64 euros, $\frac{1}{2}$ each



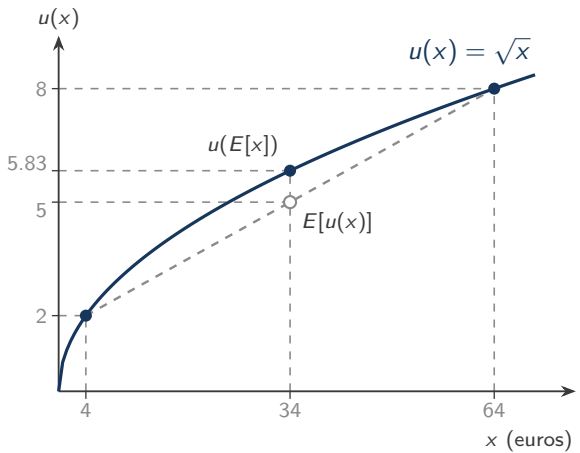
Risk averse $\Leftrightarrow$ u concave: the chord lies below the graph

Risk averse: for every F, the sure amount $\int x\, dF(x)$ is at least as good as F.

With expected utility: $\int u(x)\, dF(x) \leq u(\int x\, dF(x))$ for every F — Jensen: u concave.

Concave in money: a property of the function u, not convexity of preferences over bundles (classes 2–3). U is linear in probabilities.

MWG Def. 6.C.1 and (6.C.2), p. 185; Fig. 6.C.2, p. 186.



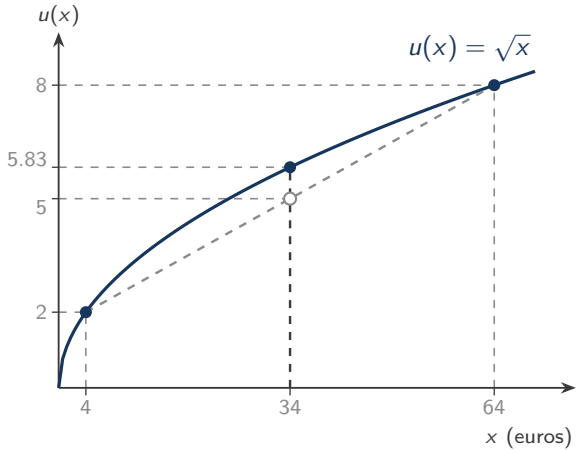
Certainty equivalent 25 euros, risk premium 9 euros: three steps

Certainty equivalent $c(F, u)$:

$u\Big(c(F,u)\Big) = \int u(x)\, dF(x)$.

Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 64 = 34$ euros



MWG Def. 6.C.2(i), (6.C.3), p. 186: "some expected return is traded for certainty". RP as in Muñoz-Garcia, p. 357.

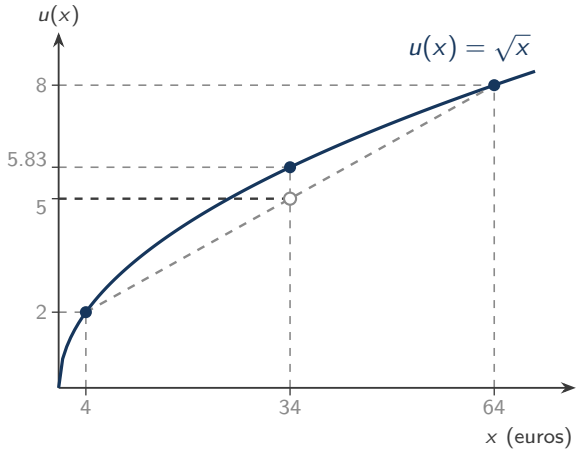
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1. $E[x] = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 8 = 5$ utils



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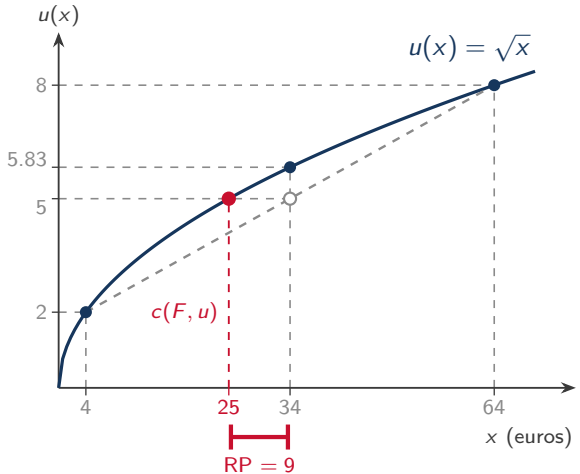
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2. $E[u(x)] = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

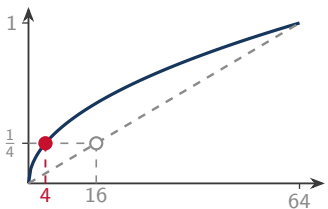
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Concave u: $c(F,u) \le E[x]$ (MWG Prop. 6.C.1); convex u: $c(F,u) \ge E[x]$

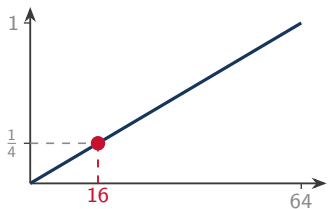
Toss two coins. Both heads: 64 euros; otherwise nothing. $E[x] = 16$ euros for everyone.

Ana: $u = \sqrt{x}/8$



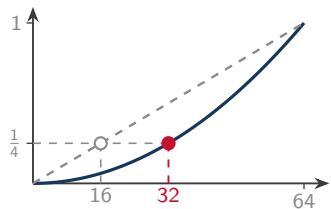
$c = 4 < 16$

Ben: $u = x/64$



$c = 16$

Carla: $u = x^2/4096$



$c = 32 > 16$

Each u rescaled so that $u(0) = 0$, $u(64) = 1$ (Prop. 6.B.2): $E[u(x)] = \frac{1}{4}$ in all three.

Risk averse $\iff$ u concave $\iff$ $c(F,u) \le E[x]$ for every F, i.e. RP ≥ 0.

MWG Prop. 6.C.1, p. 187.

Break

Back at 11:00

Practice

Five questions on Part I so far: try first, then we solve them together

Practice 1: two dice games

Prizes 150, 60, 0 euros; a lottery lists their probabilities in this order.

Iris's preferences over lotteries satisfy the independence axiom. Throw a fair die:

Game X: on 1–3, 60 euros for sure; on 4–6, the lottery $M = (0,\ \frac{1}{2},\ \frac{1}{2})$.

Game Y: on 1–3, a fair coin for 150 or 0 euros; on 4–6, M.

Iris strictly prefers Y to X.

(a) Reduce X and Y.

(b) Which must be true?

A She strictly prefers 60 euros for sure to the coin.

B She is risk loving.

C She strictly prefers the coin to 60 euros for sure.

D Nothing follows unless we know how much she values 150, 60 and 0 euros.

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(a) Reduce X and Y. $\quad X = (0,\ \frac{3}{4},\ \frac{1}{4}),\ \ Y = (\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2})$

(b) Which must be true?

A She strictly prefers 60 euros for sure to the coin.

B She is risk loving.

C She strictly prefers the coin to 60 euros for sure.

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(b) Which must be true?

A She strictly prefers 60 euros for sure to the coin.

B She is risk loving.

C She strictly prefers the coin to 60 euros for sure.

D Nothing follows unless we know how much she values 150, 60 and 0 euros.

C: independence, $\alpha = \frac{1}{2}$, common $L'' = M$. B: the coin's mean is $75 > 60$.

Practice 2: three equally likely prizes

A lottery F pays 0, 9 or 36 euros, each with probability $\frac{1}{3}$.
Rosa is an expected utility maximizer with Bernoulli utility $u(x) = \sqrt{x}$.

(a) $E[x]$ and $E[u(x)]$.

(b) Her certainty equivalent $c(F, u)$ and the risk premium $\text{RP} = E[x] - c(F, u)$.

(c) Instead of F she is offered 12 euros for sure. Does she accept? Would a risk-neutral person accept?

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(a) $E[x]$ and $E[u(x)]$.

$E[x] = 15$ euros;
$E[u(x)] = \frac{1}{3}(0 + 3 + 6) = 3$ utils

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$c(F, u) = 3^2 = 9$ euros;
$\text{RP} = 15 - 9 = 6$ euros

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$\text{RP} = 15 - 9 = 6$ euros

(c) Instead of F she is offered 12 euros for sure. Does she accept? Would a risk-neutral person accept?

Rosa: yes, $12 > 9$.
Risk neutral: no, $12 < 15$.

Practice 3: what duality says

Vera's preferences are rational, continuous, locally non-satiated and strictly convex. At prices $p = (2, 7)$, the cheapest bundle that gives her utility 5 is $h(p, 5) = (7, 4)$.

h: Hicksian demand; x: Walrasian demand.

Which statement must be true?

A $h((4, 14),\ 5) = (14, 8)$

B $x(p, 42) = (7, 4)$

C $x(p, 84) = (14, 8)$

D $x(p, 40) = (7, 4)$

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B: $(7, 4)$ costs $2 \cdot 7 + 7 \cdot 4 = 42$; with wealth 42 it is her best bundle (MWG Prop. 3.E.1). A: h does not change when all prices double. C: true only if demand is proportional to wealth. D: $(7, 4)$ costs $42 > 40$.

Practice 4: one price rise, four measures

Kim: $u(x_1, x_2) = x_1 x_2$, wealth $w = 18$.
Good 1 gets dearer: prices $p^0 = (1, 3) \to p^1 = (9, 3)$.

Class 4: $x(p, w) = \big(w/(2p_1),\ w/(2p_2)\big)$. h: Hicksian demand; e: expenditure function.

(a) $x^0 = x(p^0, w)$, u^0; $x^1 = x(p^1, w)$, u^1.

(b) The Hicksian bundle $h(p^1, u^0)$; on good 1, the substitution effect $h_1(p^1, u^0) - x_1^0$ and the wealth effect $x_1^1 - h_1(p^1, u^0)$.

(c) Compensating variation
$\text{CV} = w - e(p^1, u^0)$ and equivalent variation
$\text{EV} = e(p^0, u^1) - w$.

(d) The true cost-of-living index $e(p^1, u^0)/w$ and the fixed basket $(p^1 \cdot x^0)/(p^0 \cdot x^0)$.

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(a) $x^0 = x(p^0, w)$, u^0; $x^1 = x(p^1, w)$, u^1. $x^0 = (9, 3)$, $u^0 = 27$; $x^1 = (1, 3)$, $u^1 = 3$

(b) The Hicksian bundle $h(p^1, u^0)$; on good 1, the substitution effect $h_1(p^1, u^0) - x_1^0$ and the wealth effect $x_1^1 - h_1(p^1, u^0)$.

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$h(p^1, u^0) = (3, 9)$;
substitution $9 \to 3$: -6;
wealth $3 \to 1$: -2

(c) Compensating variation
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$\text{CV} = w - e(p^1, u^0)$ and equivalent variation
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$\text{CV} = 18 - 54 = -36$;
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(c) Compensating variation $\text{CV} = w - e(p^1, u^0)$ and equivalent variation $\text{EV} = e(p^0, u^1) - w$.	$\text{CV} = 18 - 54 = -36$; $\text{EV} = 6 - 18 = -12$
(d) The true cost-of-living index $e(p^1, u^0)/w$ and the fixed basket $(p^1 \cdot x^0)/(p^0 \cdot x^0)$.	index $54/18 = 3$; fixed basket $90/18 = 5$

Practice 5: a straight-line Bernoulli utility

Otto is an expected utility maximizer with Bernoulli utility $u(x) = 3x + 4$ over money $x \geq 0$, in euros.

Which statement is true?

A He is risk averse, since $u(0) = 4 > 0$.

B His certainty equivalent of every lottery is 4 euros above its expected value.

C He strictly prefers 10 euros for sure to a fair coin for 0 or 20 euros.

D His certainty equivalent of every lottery equals its expected value.

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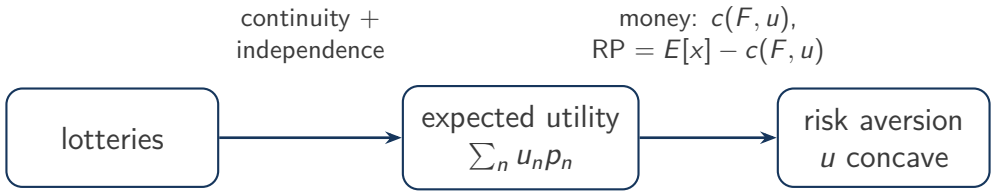
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D His certainty equivalent of every lottery equals its expected value.

D: $E[u(x)] = 3E[x] + 4 = u(E[x])$, so $c(F, u) = E[x]$: he is risk neutral. C: the coin gives $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 64 = 34 = u(10)$, so he is indifferent.

Today in one line



Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance. First- and second-order stochastic dominance, mean-preserving spreads. Review.

Read MWG 6.C–6.D (or Muñoz-Garcia ch. 5) for Tuesday 27 October.

Part I exam: Wednesday 28 October, Room 1030.
It runs from 09:30 to 11:20.